3 Lecture 3 (January 15, 2025): Affine groups and representations

Scribe: Andrew Aguilar

3.1 Recovering the Hopf structure

We finish the antiequivalence between affine group schemes and commutative Hopf algebras begun in Lecture 2.

3.1 Theorem. *The coordinate-algebra construction gives an antiequivalence between* AffGrp_k *and* $\mathsf{CommHopf}_k$.

Proof. The two constructions of the previous lecture agree with the underlying Yoneda antiequivalence. It remains to check the additional structures. Let A be a commutative Hopf algebra and let $G(R) = \mathrm{Hom}_{k\text{-}\mathsf{Alg}}(A, R)$ with its convolution product. The two algebra maps

$$\iota_l(a) = a \otimes 1, \qquad \iota_r(a) = 1 \otimes a$$

are points of $G(A \otimes A)$. Their product is

$$(\iota_l \iota_r)(a) = \sum (a_1 \otimes 1)(1 \otimes a_2) = \sum a_1 \otimes a_2 = \Delta(a).$$

Thus the group law recovers the original comultiplication. The identity of $G(k)$ is $\epsilon : A \to k$, and the inverse of the universal point $\mathrm{id}_A \in G(A)$ is $S : A \to A$. This recovers the counit and antipode. Conversely, Yoneda identifies the natural group operations on any affine G with precisely these three algebra maps, so they are recovered as well. Finally, a natural map of affine schemes is a group homomorphism exactly when its coordinate-algebra map commutes with these operations. Hence the constructions are inverse on objects and morphisms, up to their natural Yoneda identifications. ∎

For clarity, the multiplication on $A \otimes B$ used above is

$$(a \otimes b)(a' \otimes b') = aa' \otimes bb'.$$

It is the composite of $\mathrm{id}_A \otimes \tau_{B,A} \otimes \mathrm{id}_B$ with $m_A \otimes m_B$, where $\tau_{B,A}(b \otimes a') = a' \otimes b$.

3.2 Example. The additive group $\mathbb{G}_a$ has coordinate algebra $k[x]$, with

$$\Delta(x) = x \otimes 1 + 1 \otimes x, \qquad \epsilon(x) = 0, \qquad S(x) = -x.$$

These formulas express addition, zero and negation.

3.3 Example. For GL_n, put $X = (x_{ij})$ and $d = \det X$. Its coordinate algebra is $A = k[x_{ij}, d^{-1}]$. Matrix multiplication gives

$$\Delta(x_{ij}) = \sum_{l=1}^{n} x_{il} \otimes x_{lj}, \qquad \Delta(d^{-1}) = d^{-1} \otimes d^{-1}.$$

The identity matrix and inverse matrix give

$$\epsilon(x_{ij}) = \delta_{ij}, \qquad S(x_{ij}) = (X^{-1})_{ij}, \qquad \epsilon(d^{-1}) = 1, \quad S(d^{-1}) = d.$$

3.4 Exercise. Check directly that these formulas respect $dd^{-1} = 1$ and satisfy the counit and antipode identities.

3.5 Example. For $\mathbb{G}_m = \mathrm{GL}_1$ the algebra is $k[x, x^{-1}]$, and

$$\Delta(x) = x \otimes x, \qquad \epsilon(x) = 1, \qquad S(x) = x^{-1}.$$

3.2 Rational representations and comodules

Fix a field k, an affine group scheme G and a k-vector space V. Write $V_R = V \otimes_k R$ and $\mathrm{GL}(V)(R) = \mathrm{Aut}_R(V_R)$. When V is finite-dimensional, the functor $R \mapsto V_R$ is an affine scheme represented by $\mathrm{Sym}(V^*)$, and $\mathrm{GL}(V)$ is an affine group scheme isomorphic to $\mathrm{GL}_{\dim V}$ after choosing a basis. For arbitrary V, we use these functors without asserting that they are affine schemes.

3.6 Definition. A *rational G-module* is a vector space V with an action $G(R) \curvearrowright V_R$ by R-linear automorphisms, natural in every commutative k-algebra R. Equivalently, it is a morphism of group-valued functors $G \to \mathrm{GL}(V)$.

3.7 Definition. Let (A, Δ, ϵ) be a coalgebra over k. A *right A-comodule* is a vector space M with a linear map $\rho : M \to M \otimes A$ such that

$$(\rho \otimes \mathrm{id}_A)\rho = (\mathrm{id}_M \otimes \Delta)\rho, \qquad (\mathrm{id}_M \otimes \epsilon)\rho = \mathrm{id}_M .$$

These are coassociativity and counitality. In Sweedler notation we write $\rho(m) = \sum m_0 \otimes m_1$, with the summation understood to be finite for each m.

3.8 Theorem. *Let $A = k[G]$. Giving a rational G-module structure on V is equivalent to giving a right A-comodule structure on V. If V is finite-dimensional, these are also equivalent to a morphism of affine group schemes $G \to \mathrm{GL}(V)$.*

We prove this statement and its compatibility with morphisms in the next lecture.

References

[BEO20] Dave Benson, Pavel Etingof, and Victor Ostrik. New incompressible symmetric tensor categories in positive characteristic, 2020.

[CEO23] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. On Frobenius exact symmetric tensor categories. *Annals of Mathematics*, 197(3):1235–1279, 2023.

[CEO24] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. Incompressible tensor categories. *Advances in Mathematics*, 457:109935, 2024.

[CF24] Kevin Coulembier and Johannes Flake. Towards higher Frobenius functors for symmetric tensor categories, 2024. The course discussion follows the May 2024 version.

[Del02] Pierre Deligne. Catégories tensorielles. *Moscow Mathematical Journal*, 2(2):227–248, 2002.

[DM82] Pierre Deligne and James S. Milne. Tannakian categories. In *Hodge Cycles, Motives, and Shimura Varieties*, volume 900 of *Lecture Notes in Mathematics*, pages 101–228. Springer, 1982. Corrected version, 2022, with the original proposition numbering.

[EGNO15] Pavel Etingof, Shlomo Gelaki, Dmitri Nikshych, and Victor Ostrik. *Tensor categories*, volume 205 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2015.

[ENO05] Pavel Etingof, Dmitri Nikshych, and Viktor Ostrik. On fusion categories. *Annals of Mathematics*, 162(2):581–642, 2005.

[EO19] Pavel Etingof and Victor Ostrik. On semisimplification of tensor categories, 2019.

[Mil12] James S. Milne. Basic theory of affine group schemes, 2012. Available at www.jmilne.org/math/.

[Ost15] Victor Ostrik. On symmetric fusion categories in positive characteristic, 2015.

[Wat79] William. C. Waterhouse. *Introduction to Affine Group Schemes*. Springer Graduate texts in mathematics. Springer New York, NY, 1979.

[Wib25] Michael Wibmer. A remark on torsors under affine group schemes. *Transformation Groups*, 30:447–454, 2025.